1. Create a directory named archive_lab and create three text files inside it. Add sample content to each file.

```
akansha@localhost:~$ mkdir archive_lab
akansha@localhost:~$ echo "Linux Archiving Lab 11 - File 1 Content" >
archive_lab/file1.txt
akansha@localhost:~$ echo "Linux Archiving Lab 11 - File 2 Content" >
archive_lab/file2.txt
akansha@localhost:~$ echo "Linux Archiving Lab 11 - File 3 Content" >
archive_lab/file3.txt
akansha@localhost:~$ ls -l archive_lab
total 12
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:50 file1.txt
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:50 file2.txt
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:50 file3.txt
```

2. Create a tar archive named files.tar containing all files from archive_lab. Verify that the archive was created.

```
akansha@localhost:~$ tar -cvf files.tar archive_lab/
archive_lab/
archive_lab/file1.txt
archive_lab/file2.txt
archive_lab/file3.txt
akansha@localhost:~$ ls -l files.tar
-rw-r--r--. 1 akansha akansha 10240 Oct 31 20:51 files.tar
```

3. List the contents of files.tar without extracting it.

```
akansha@localhost:~$ tar -tvf files.tar
drwxr-xr-x akansha/akansha     0 2026-10-31 20:50 archive_lab/
-rw-r--r-- akansha/akansha    43 2026-10-31 20:50 archive_lab/file1.txt
-rw-r--r-- akansha/akansha    43 2026-10-31 20:50 archive_lab/file2.txt
-rw-r--r-- akansha/akansha    43 2026-10-31 20:50 archive_lab/file3.txt
```

4. Extract files.tar into a new directory named extracted_tar.

```
akansha@localhost:~$ mkdir extracted_tar
akansha@localhost:~$ tar -xvf files.tar -C extracted_tar/
archive_lab/
archive_lab/file1.txt
archive_lab/file2.txt
archive_lab/file3.txt
akansha@localhost:~$ ls -l extracted_tar/archive_lab
total 12
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:50 file1.txt
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:50 file2.txt
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:50 file3.txt
```

5. Create a gzip-compressed file from a text file and observe the change in its filename.

```
akansha@localhost:~$ cp archive_lab/file1.txt ./single_test.txt
akansha@localhost:~$ ls -l single_test.txt
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:52 single_test.txt
akansha@localhost:~$ gzip single_test.txt
akansha@localhost:~$ ls -l single_test.txt.gz
-rw-r--r--. 1 akansha akansha 65 Oct 31 20:52 single_test.txt.gz
```

6. Use gunzip to decompress the gzip-compressed file and verify that the original file is restored.

```
akansha@localhost:~$ gunzip single_test.txt.gz
akansha@localhost:~$ ls -l single_test.txt
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:52 single_test.txt
akansha@localhost:~$ cat single_test.txt
Linux Archiving Lab 11 - File 1 Content
```

7. Create a tar.gz archive named archive.tar.gz containing the complete archive_lab directory.

```
akansha@localhost:~$ tar -czvf archive.tar.gz archive_lab/
archive_lab/
archive_lab/file1.txt
archive_lab/file2.txt
archive_lab/file3.txt
akansha@localhost:~$ ls -l archive.tar.gz
-rw-r--r--. 1 akansha akansha 175 Oct 31 20:53 archive.tar.gz
```

8. List the contents of archive.tar.gz without extracting it.

```
akansha@localhost:~$ tar -tzvf archive.tar.gz
drwxr-xr-x akansha/akansha     0 2026-10-31 20:50 archive_lab/
-rw-r--r-- akansha/akansha    43 2026-10-31 20:50 archive_lab/file1.txt
-rw-r--r-- akansha/akansha    43 2026-10-31 20:50 archive_lab/file2.txt
-rw-r--r-- akansha/akansha    43 2026-10-31 20:50 archive_lab/file3.txt
```

9. Extract archive.tar.gz into a separate directory and verify the extracted files.

```
akansha@localhost:~$ mkdir extracted_targz
akansha@localhost:~$ tar -xzvf archive.tar.gz -C extracted_targz/
archive_lab/
archive_lab/file1.txt
archive_lab/file2.txt
archive_lab/file3.txt
akansha@localhost:~$ ls -l extracted_targz/archive_lab
total 12
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:50 file1.txt
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:50 file2.txt
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:50 file3.txt
```

10. Create a zip archive named files.zip containing three selected files.

```
akansha@localhost:~$ zip files.zip archive_lab/file1.txt archive_lab/file2.txt
archive_lab/file3.txt
  adding: archive_lab/file1.txt (deflated 12%)
  adding: archive_lab/file2.txt (deflated 12%)
  adding: archive_lab/file3.txt (deflated 12%)
akansha@localhost:~$ ls -l files.zip
-rw-r--r--. 1 akansha akansha 512 Oct 31 20:54 files.zip
```

11. List the contents of files.zip without extracting it.

```
akansha@localhost:~$ unzip -l files.zip
Archive:  files.zip
  Length      Date    Time    Name
---------  ---------- -----   ----
       43  10-31-2026 20:50   archive_lab/file1.txt
       43  10-31-2026 20:50   archive_lab/file2.txt
       43  10-31-2026 20:50   archive_lab/file3.txt
---------                     -------
      129                     3 files
```

12. Extract files.zip into a directory named extracted_zip.

```
akansha@localhost:~$ unzip files.zip -d extracted_zip
Archive:  files.zip
  inflating: extracted_zip/archive_lab/file1.txt
  inflating: extracted_zip/archive_lab/file2.txt
  inflating: extracted_zip/archive_lab/file3.txt
akansha@localhost:~$ ls -l extracted_zip/archive_lab
total 12
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:50 file1.txt
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:50 file2.txt
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:50 file3.txt
```

13. Use unzip to extract only one selected file from a ZIP archive.

```
akansha@localhost:~$ mkdir single_extract
akansha@localhost:~$ unzip files.zip archive_lab/file1.txt -d single_extract
Archive:  files.zip
  inflating: single_extract/archive_lab/file1.txt
akansha@localhost:~$ ls -l single_extract/archive_lab
total 4
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:50 file1.txt
```

14. Create a ZIP archive containing an entire directory and verify its contents.

```
akansha@localhost:~$ zip -r full_directory.zip archive_lab/
  adding: archive_lab/ (stored 0%)
  adding: archive_lab/file1.txt (deflated 12%)
  adding: archive_lab/file2.txt (deflated 12%)
  adding: archive_lab/file3.txt (deflated 12%)
akansha@localhost:~$ unzip -l full_directory.zip
Archive:  full_directory.zip
  Length      Date    Time    Name
---------  ---------- -----   ----
        0  10-31-2026 20:50   archive_lab/
       43  10-31-2026 20:50   archive_lab/file1.txt
       43  10-31-2026 20:50   archive_lab/file2.txt
       43  10-31-2026 20:50   archive_lab/file3.txt
---------                     -------
      129                     4 files
```

15. Create a gzip-compressed tar archive and explain the difference between tar and gzip in the command sequence used.

```
akansha@localhost:~$ tar -cvf archive.tar archive_lab/
archive_lab/
archive_lab/file1.txt
archive_lab/file2.txt
archive_lab/file3.txt
akansha@localhost:~$ gzip archive.tar
akansha@localhost:~$ ls -l archive.tar.gz
-rw-r--r--. 1 akansha akansha 175 Oct 31 20:55 archive.tar.gz

Explanation:
- tar (Tape Archiver) binds multiple files and directories into a single archive
file without compression.
- gzip (GNU zip) compresses a single file to reduce its size.
- Using tar -czvf combines both steps simultaneously (archiving with tar and
compressing with gzip).
```

16. Create a tar archive, add another file to the archive, and verify that the new file is present.

```
akansha@localhost:~$ echo "Linux Lab 11 Extra File" > extra.txt
akansha@localhost:~$ tar -cvf append_test.tar archive_lab/
archive_lab/
archive_lab/file1.txt
archive_lab/file2.txt
archive_lab/file3.txt
akansha@localhost:~$ tar -rvf append_test.tar extra.txt
extra.txt
akansha@localhost:~$ tar -tvf append_test.tar
drwxr-xr-x akansha/akansha     0 2026-10-31 20:50 archive_lab/
-rw-r--r-- akansha/akansha    43 2026-10-31 20:50 archive_lab/file1.txt
-rw-r--r-- akansha/akansha    43 2026-10-31 20:50 archive_lab/file2.txt
-rw-r--r-- akansha/akansha    43 2026-10-31 20:50 archive_lab/file3.txt
-rw-r--r-- akansha/akansha    24 2026-10-31 20:56 extra.txt
```

17. Compare the archive contents and file sizes of a tar archive and a compressed tar.gz archive.

```
akansha@localhost:~$ ls -lh files.tar archive.tar.gz
-rw-r--r--. 1 akansha akansha 175 Oct 31 20:53 archive.tar.gz
-rw-r--r--. 1 akansha akansha 10K Oct 31 20:51 files.tar

Comparison Analysis:
- Uncompressed tar (.tar) keeps exact block padding structure (10 KB total).
- Compressed archive (.tar.gz) uses DEFLATE algorithm reducing total file
footprint to 175 Bytes (~98% reduction).
```

18. Use appropriate commands to decompress and extract a tar.gz archive and a ZIP archive into separate directories.

```
akansha@localhost:~$ mkdir -p restored_targz restored_zip
akansha@localhost:~$ tar -xzvf archive.tar.gz -C restored_targz/
archive_lab/
archive_lab/file1.txt
archive_lab/file2.txt
archive_lab/file3.txt
akansha@localhost:~$ unzip full_directory.zip -d restored_zip/
Archive:  full_directory.zip
   creating: restored_zip/archive_lab/
  inflating: restored_zip/archive_lab/file1.txt
  inflating: restored_zip/archive_lab/file2.txt
  inflating: restored_zip/archive_lab/file3.txt
```

19. Create a complete backup archive of a practice directory using tar.gz, then restore it into another directory.

```
akansha@localhost:~$ tar -czvf system_backup.tar.gz archive_lab/
archive_lab/
archive_lab/file1.txt
archive_lab/file2.txt
archive_lab/file3.txt
akansha@localhost:~$ mkdir -p /tmp/backup_restore
akansha@localhost:~$ tar -xzvf system_backup.tar.gz -C /tmp/backup_restore/
archive_lab/
archive_lab/file1.txt
archive_lab/file2.txt
archive_lab/file3.txt
akansha@localhost:~$ ls -l /tmp/backup_restore/archive_lab
total 12
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:50 file1.txt
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:50 file2.txt
-rw-r--r--. 1 akansha akansha 43 Oct 31 20:50 file3.txt
```

20. Demonstrate a complete archive and compression workflow using tar, gzip, gunzip, zip, and unzip, including verification after each operation.

```
akansha@localhost:~$ tar -czvf workflow_demo.tar.gz archive_lab/ && tar -tzvf
workflow_demo.tar.gz
archive_lab/
archive_lab/file1.txt
archive_lab/file2.txt
archive_lab/file3.txt
drwxr-xr-x akansha/akansha     0 2026-10-31 20:50 archive_lab/
-rw-r--r-- akansha/akansha    43 2026-10-31 20:50 archive_lab/file1.txt
-rw-r--r-- akansha/akansha    43 2026-10-31 20:50 archive_lab/file2.txt
-rw-r--r-- akansha/akansha    43 2026-10-31 20:50 archive_lab/file3.txt

akansha@localhost:~$ zip -r workflow_demo.zip archive_lab/ && unzip -l
workflow_demo.zip
  adding: archive_lab/ (stored 0%)
  adding: archive_lab/file1.txt (deflated 12%)
  adding: archive_lab/file2.txt (deflated 12%)
  adding: archive_lab/file3.txt (deflated 12%)
Archive:  workflow_demo.zip
  Length      Date    Time    Name
---------  ---------- -----   ----
        0  10-31-2026 20:50   archive_lab/
       43  10-31-2026 20:50   archive_lab/file1.txt
       43  10-31-2026 20:50   archive_lab/file2.txt
       43  10-31-2026 20:50   archive_lab/file3.txt
---------                     -------
      129                     4 files

akansha@localhost:~$ rm -rf archive_lab files.tar single_test.txt*
archive.tar.gz extracted_* files.zip single_extract full_directory.zip extra.txt
append_test.tar restored_* system_backup.tar.gz /tmp/backup_restore
workflow_demo.*
akansha@localhost:~$ echo "Linux Lab 11 Archiving & Compression workflow
completed and cleaned."
Linux Lab 11 Archiving & Compression workflow completed and cleaned.
```